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CHAPTER 1

INTRODUCTION

India is home to approximately 146 million farming households, the majority of which are smallholder farmers cultivating less than two hectares of land. These farmers form the backbone of India's agricultural ecosystem and collectively contribute nearly 51 percent of the nation's total food production. Despite their immense contribution to national food security, smallholder farmers continue to operate with limited access to real-time market intelligence, modern agricultural advisory systems, government welfare schemes, crop insurance, and emerging financial instruments such as carbon finance. This persistent lack of integrated support significantly reduces their productivity, profitability, and resilience.

The agricultural sector in India contributes approximately 17 percent to the national Gross Domestic Product (GDP) and employs more than 42 percent of the country's workforce, making it one of the most critical pillars of the Indian economy. However, farmer income remains deeply inconsistent due to several systemic challenges, including information asymmetry, lack of direct market access, climate uncertainty, delayed crop disease detection, and the absence of digital tools specifically designed for the rural Indian context. While smartphone adoption has increased significantly even in semi-rural and rural regions, existing agricultural platforms remain fragmented, offering isolated services such as market access, weather forecasting, or advisory support without consolidating them into a single unified ecosystem. As a result, there remains no comprehensive, Hindi-first, mobile-first platform that addresses the full spectrum of a farmer's operational, financial, and sustainability needs.

Krishi-Drishti, meaning "Farmer Vision" in Hindi, is proposed as a comprehensive AI-powered agricultural intelligence platform developed to bridge this critical gap. The platform combines Artificial Intelligence (AI), satellite remote sensing, machine learning, predictive analytics, and real-time API integrations into a single multilingual digital interface available in both Hindi and English. Designed specifically for smallholder and marginal farmers, Krishi-Drishti aims to function as a complete digital agricultural companion that empowers users through informed decision-making, direct market participation, sustainability incentives, and climate resilience.

The platform provides a wide range of integrated functionalities that collectively transform fragmented agricultural services into one intelligent ecosystem. These include a direct agricultural marketplace (Mandi) that enables farmers to list produce and connect directly with buyers, thereby bypassing exploitative middlemen and improving price transparency. It also incorporates satellite-based crop health monitoring using NDVI (Normalized Difference Vegetation Index) and EVI (Enhanced Vegetation Index) metrics derived from Google Earth Engine, enabling real-time crop surveillance and precision agriculture insights at plot level.

Krishi-Drishti further integrates AI-powered plant disease diagnosis through Google Gemini Vision, allowing farmers to identify pathogens, diseases, or nutrient deficiencies directly from

crop images. In addition, the platform introduces a Carbon Vault module that tracks sustainable agricultural practices and enables enrollment into verified carbon credit ecosystems, thereby connecting smallholder farmers with climate finance opportunities previously inaccessible to them.

To improve institutional accessibility, the platform includes Scheme Setu, a government welfare scheme discovery and digital application portal tailored to farmer profiles, ensuring that users can identify and access relevant subsidies, insurance programs, and support services. The system also offers real-time weather forecasting with a seven-day outlook, disease-risk alerts, and predictive epidemiological support for climate-smart decision-making.

A unique innovation of Krishi-Drishti is its bioacoustic analysis capability, which utilizes smartphone microphones to detect pests or crop diseases early through canopy vibration and sound pattern analysis. Complementing this is an AI-powered Soil Carbon Model that estimates Soil Organic Carbon (SOC) using satellite-derived metrics and predictive machine learning, enabling farmers to optimize soil health while participating in sustainability-linked agricultural systems.

By integrating market intelligence, AI diagnostics, satellite surveillance, sustainability finance, multilingual advisory, and government inclusion into one mobile-first platform, Krishi-Drishti seeks to redefine the digital agricultural landscape for Indian farmers. The platform is not merely a technological solution but a comprehensive empowerment ecosystem designed to increase farmer income, reduce agricultural risk, improve productivity, and support the transition toward climate-resilient and data-driven farming.

The platform serves as a digital companion for Indian farmers by providing:

- Direct agricultural marketplace (Mandi) for farmers to sell produce directly to buyers without middlemen.
- Satellite-based crop health monitoring using NDVI and EVI through Google Earth Engine.
- AI-powered plant disease diagnosis using Google Gemini Vision from crop images.
- Carbon Vault for sustainable farming tracking and carbon credit enrollment.
- Scheme Setu for government scheme discovery and digital application support.
- Real-time 7-day weather forecasting with disease-risk alerts.
- Bioacoustic pest and disease detection using smartphone microphone analysis.
- AI-powered Soil Carbon Model for Soil Organic Carbon estimation using satellite data.

KEY OBJECTIVES :

Figure 1.1 (project objectives)



Predictive Agricultural Intelligence:

To develop a highly accurate and resilient predictive intelligence system that leverages advanced Stacking ensemble methodologies for stable Soil Organic Carbon (SOC) estimation, crop health forecasting, and environmental risk analysis. By overcoming the limitations of traditional machine learning models—such as overfitting, poor generalization, and inconsistent field-level performance—the platform aims to deliver scientifically reliable predictions that support precision agriculture and long-term soil sustainability.

Proactive Disease Prevention and Early Intervention:

To transform agricultural monitoring from a reactive process into a proactive defense system by integrating AI-driven disease surveillance, satellite change detection, and GenAI-powered advisory services. Rather than simply identifying crop issues after damage occurs, the system is designed to provide early-warning alerts, predictive disease-risk assessments, and preventive recommendations—such as fungicide scheduling, irrigation correction, or nutrient interventions—directly to farmers through an intuitive multilingual chatbot interface.

Sustainability Incentivization through Carbon Intelligence:

To establish a digitally verified sustainability ecosystem by incorporating a Tokenized Internal Carbon Credit framework that rewards farmers for adopting regenerative and carbon-positive agricultural practices. Through blockchain-backed carbon accounting, SOC improvement

tracking, and sustainable land-use monitoring, the platform seeks to create a self-optimizing economic loop where environmentally responsible farming directly translates into measurable financial incentives.

Disease Mitigation and Crop Loss Reduction:

To significantly reduce agricultural losses by deploying continuous crop surveillance systems that combine remote sensing, AI-based disease diagnostics, and anomaly detection models for early outbreak identification. This objective focuses on minimizing the economic and ecological impact of pests, pathogens, and environmental stress through timely intervention and data-driven risk management.

Soil Optimization and Nutrient Management:

To provide farmers with dependable, real-time soil intelligence by enabling accurate monitoring of Soil Organic Carbon, vegetation health, and soil productivity indicators. This objective supports smarter fertilizer application, nutrient balancing, and land restoration strategies, thereby improving both agricultural productivity and ecological resilience.

Farmer Empowerment through Accessible AI:

To democratize access to expert-level agricultural knowledge by designing an inclusive, user-friendly AI ecosystem that translates complex scientific analysis into practical, understandable guidance for farmers. Through multilingual interfaces, conversational AI, and mobile-first accessibility, the platform ensures that even smallholder and semi-literate farmers can benefit from advanced agricultural intelligence.

Climate-Resilient and Carbon-Conscious Agriculture:

To accelerate the transition toward climate-smart farming by integrating digital ledger systems, carbon tracking, and sustainability analytics into everyday agricultural decision-making. This objective promotes environmentally responsible farming practices that align economic productivity with national and global climate action goals.

Integrated Rural Digital Transformation:

To build a unified agricultural intelligence ecosystem that connects crop monitoring, disease diagnosis, market access, financial inclusion, government schemes, and carbon finance into a single digital framework—ultimately transforming fragmented agricultural support systems into a comprehensive farmer-centric platform for rural empowerment.

CHAPTER 2

LITERATURE SURVEY

Table 2.1 (Comparative Literature Survey of Existing Research on Smart Agriculture)

Title	Authors	Publication Year	Outcomes	Drawbacks
Automated Applications of Acoustics for Stored Product Insect Detection, Monitoring, and Management	Mankin, R.W., et al.	2021	Demonstrated successful detection of hidden insect infestations such as larvae inside grains using digital signal processing and machine learning classifiers. Proved that bioacoustic systems can detect pests often missed by traditional visual inspection methods, significantly improving early pest surveillance and stored grain protection.	High hardware and deployment costs , requiring specialized acoustic sensors and recording equipment; computational complexity for real-time edge processing; limited affordability for smallholder farmers ; difficulty in noisy open-field environments; and lack of integration with broader precision agriculture, advisory, or farm management systems.
Image-based Crop Disease Detection with Federated Learning	Mamba Kabala, D., Hafiane, A., Bobelin, L., Canals, R.	2023	Developed a privacy-preserving federated learning framework for crop disease detection using CNN architectures such as ResNet50 and MobileNetV2. Reduced centralized data dependency, improved farmer data privacy, and demonstrated that CNNs outperform Vision Transformers	Diagnostic-only focus without treatment recommendations or preventive guidance; computationally demanding for low-resource smartphones; dependence on image quality and internet synchronization; challenges in rural deployment; and absence of integration with weather, market access, carbon finance, or comprehensive

Title	Authors	Publication Year	Outcomes	Drawbacks
			in decentralized agricultural environments.	agricultural intelligence systems.
Remote Sensing Based Crop Health Classification Using NDVI and Fully Connected Neural Networks	Judith, J., Tamilselvi, R., Beham, M.P., Lakshmi, S.S.P., Panthakkan, A., Mansoori, S.A., Ahmad, H.A.	2024	Achieved 97.80% crop health classification accuracy using NDVI, GNDVI, EVI, MSAVI, and Hybrid Vegetation Index with Fully Connected Neural Networks, proving superior predictive performance over traditional linear models for crop stress analysis.	Relies solely on satellite spectral data , making it incapable of identifying exact stress causes such as nutrient deficiency, pests, disease, or irrigation issues; no ground-level contextual validation ; weather/cloud dependency; and lack of socioeconomic or practical advisory integration for actionable farmer decision-making.
Agro-Blockchain Markets (ABM): A Secure and Transparent Model for Smallholder Farmers	Ajayi, O.O., Toromade, A.S., Olagoke,			

2.1 Existing Systems Analysis

2.1.1 AgriStack and PM-KISAN Digital Infrastructure

The Government of India's AgriStack initiative provides foundational digital infrastructure for the agricultural sector, including Farmer Databases, Soil Health Cards, and digitized administrative records. These systems are beneficial for governance and policy-level digitization; however, they primarily function as administrative frameworks and lack interactive farmer advisory, direct market connectivity, AI-powered crop disease diagnosis, and real-time precision agriculture capabilities. As a result, while AgriStack strengthens

backend agricultural data systems, it does not serve as a comprehensive farmer-centric operational platform.

2.1.2 Agrimarket and eNAM

eNAM (National Agriculture Market) is a pan-India electronic trading platform designed to improve agricultural trade through digital market access. It enables farmers to connect with broader mandi systems and buyers across regions. However, the platform requires significant smartphone proficiency and digital literacy, while also depending heavily on stable internet connectivity within mandi ecosystems—an issue in many rural areas. Additionally, eNAM lacks integrated AI advisory, predictive disease analytics, personalized recommendations, and carbon credit functionalities, limiting its role to market linkage rather than holistic agricultural empowerment.

2.1.3 AgroStar and Kisan Suvidha

AgroStar and Kisan Suvidha offer agronomy guidance, weather information, and agricultural input purchasing support. These platforms have improved accessibility to basic farming recommendations and agri-commerce services. However, they remain largely one-dimensional by focusing on advisory and input access without integrating advanced satellite monitoring, conversational AI, predictive intelligence, or sustainability frameworks such as carbon finance. Consequently, their functionality is limited when compared to modern integrated agricultural intelligence ecosystems.

2.1.4 Boomitra Carbon Platform

Boomitra is a carbon-focused agricultural platform that enables sustainability participation through carbon credit systems. However, it operates largely on an aggregator-only model that creates accessibility challenges for Hindi-speaking and regional farmers, while the onboarding process—typically lasting 30 to 45 minutes—can be lengthy and complex for digitally inexperienced users. These factors reduce inclusivity among India's rural s

smallholder population.

2.1.5 Grow Indigo

Grow Indigo primarily targets organized and relatively large-scale agricultural operations with sustainability initiatives. While valuable in carbon and regenerative agriculture, its penetration among smallholder farmers remains limited. The platform is not strongly mobile-first, lacks localized conversational accessibility, and does not provide real-time satellite-driven AI advisory tailored to individual plot-level decision-making for smaller rural users.

2.1.6 EKI Energy Services

EKI Energy Services supports carbon credit validation and sustainability frameworks through established methodologies. However, it relies heavily on manual and expensive third-party audits, making the process administratively burdensome and slow. Credit verification can take several months, and farmers often lack direct visibility into carbon credit generation, earnings, or real-time status. This opacity and complexity significantly reduce accessibility for small and marginal farmers.

2.1.7 Summary of Gaps in Existing Systems

- Lack of a unified platform integrating market access, AI advisory, satellite intelligence, disease diagnosis, and carbon credit systems.
 - Absence of truly Hindi-first and mobile-first design optimized for rural and semi-literate farming communities.
 - Carbon credit ecosystems remain opaque, intermediary-dependent, and difficult for smallholders to access directly.
 - Satellite intelligence and remote sensing remain technically inaccessible to most individual farmers.
 - Existing systems are fragmented, solving isolated agricultural problems without providing an end-to-end digital farming ecosystem.
-

2.2 Proposed System

2.2.1 Multilingual AI Advisory

The proposed system introduces a conversational AI-powered advisory module using Google Gemini that provides scientifically grounded, real-time agricultural guidance in both Hindi and English. This multilingual interface ensures broader accessibility and democratizes expert agricultural knowledge for smallholder farmers.

2.2.2 Direct-to-Farmer Digital Marketplace

Krishi-Drishti includes a Mandi module that allows farmers to list produce directly for buyers, reducing dependency on commission agents (arhatiyas), improving price transparency, and increasing farmer profitability through direct market participation.

2.2.3 Satellite-Based Farm Intelligence

The platform integrates Google Earth Engine to automatically retrieve and analyze vegetation health metrics such as NDVI, EVI, and MSAVI for individual plot polygons. This enables precision agriculture, real-time crop monitoring, and proactive farm intelligence at affordable scale.

2.2.4 AI Plant Disease Vision Diagnosis

Using Gemini Pro Vision, the system enables farmers to upload crop images for AI-powered disease and pathogen identification. This facilitates early diagnosis, treatment recommendations, and reduced crop loss through visual intelligence.

2.2.5 Carbon Credit Enrollment and Tracking

Krishi-Drishti incorporates a transparent Carbon Vault dashboard aligned with Verra VM0042 standards to track baseline emissions, carbon sequestration, and farmer-eligible carbon credits. This simplifies climate finance participation for smallholder farmers.

2.2.6 Scheme Setu

Scheme Setu is a personalized government welfare aggregator that recommends relevant subsidies, schemes, and insurance programs based on farmer-specific attributes such as landholding size, crop type, and district.

2.2.7 Disease Risk Forecasting

The system employs predictive epidemiological models powered by weather history, environmental conditions, and Open-Meteo integrations to forecast disease outbreaks such as powdery mildew, leaf blight, and fungal infections before major damage occurs.

2.2.8 Bioacoustic Pest Detection

An experimental bioacoustic intelligence module analyzes soil and canopy sound frequencies through mobile microphones to identify early pest infestations or crop anomalies, creating a novel preventive monitoring mechanism.

Table 2.2 (Comparative Gap Analysis of Existing Systems)

Platform/System	Market Access	AI Advisory	Satellite Monitoring	Carbon Credit Integration	Hindi-First Accessibility
AgriStack / PM-KISAN	Limited	No	No	No	Partial
eNAM	Yes	No	No	No	Partial
AgroStar / Kisan Suvidha	Limited	Basic	No	No	Partial

Platform/System	Market Access	AI Advisory	Satellite Monitoring	Carbon Credit Integration	Hindi-First Accessibility
Boomitra	No	No	Limited	Yes	No
Grow Indigo	Limited	No	Limited	Yes	Limited
EKI Energy Services	No	No	No	Yes	No
Krishi-Drishti (Proposed)	Yes	Advanced GenAI	Yes	Yes	Yes

2.3 Problem Statement

Indian agriculture, despite being the backbone of the national economy and supporting millions of livelihoods, continues to face critical structural, technological, and socioeconomic challenges that disproportionately affect smallholder farmers. The majority of Indian farmers cultivate less than two hectares of land and often operate with limited access to real-time agricultural intelligence, modern technology, transparent marketplaces, and sustainability-linked financial opportunities. These gaps reduce productivity, increase vulnerability, and prevent farmers from fully participating in digital and climate-resilient agricultural ecosystems. The following key problem areas define the major challenges addressed by the proposed Krishi-Drishti platform.

P1 – Lack of Timely Market Information

A major challenge faced by smallholder farmers is the absence of timely and transparent market intelligence, which results in income losses estimated between 20–40% due to information asymmetry, exploitative intermediaries, and limited awareness of fair market prices. Without access to real-time mandi pricing, direct buyer connectivity, or digital agricultural marketplaces, farmers are frequently compelled to sell produce below fair market value. This significantly reduces profitability, weakens bargaining power, and creates long-term financial instability.

P2 – Delayed Crop Disease Detection

Crop productivity is severely affected by delayed disease identification, with farmers experiencing yield losses ranging from 10–50% because crop diseases, pest infestations, or nutrient deficiencies are often recognized only after visible symptoms become severe. Traditional approaches depend heavily on manual field inspection or delayed expert consultation, preventing timely intervention. This delay increases crop damage, treatment costs, pesticide misuse, and overall agricultural risk.

P3 – Absence of Real-Time Farm Monitoring

Smallholder farmers often lack access to affordable real-time farm monitoring technologies such as satellite crop surveillance, vegetation health analysis, and predictive precision agriculture tools. While advanced technologies like NDVI, EVI, and remote sensing can provide valuable crop intelligence, high complexity, technical barriers, and cost constraints prevent widespread adoption among individual farmers. As a result, many farmers remain dependent on reactive decision-making instead of proactive, data-driven farm management strategies.

P4 – Carbon Market Exclusion

Despite the increasing importance of climate-smart agriculture and carbon finance, most smallholder farmers remain excluded from carbon credit ecosystems due to complex verification systems, technical literacy barriers, administrative burdens, and inaccessible sustainability platforms. This exclusion prevents farmers from monetizing regenerative agricultural practices such as carbon sequestration, conservation agriculture, and sustainable soil management, thereby limiting both environmental and economic opportunities.

P5 – Poor Government Scheme Awareness

A substantial proportion of eligible farmers—estimated at over 40%—remain unaware of critical government welfare schemes, agricultural subsidies, crop insurance programs, and digital support services due to fragmented communication systems, language barriers, and procedural complexity. This lack of awareness and accessibility prevents farmers from leveraging institutional support systems that could improve financial security, agricultural resilience, and social welfare.

P6 – Climate Vulnerability

Farmers are increasingly vulnerable to unpredictable climate conditions such as droughts, floods, pest outbreaks, irregular rainfall, and seasonal instability. However, many lack access to preventive advisory systems capable of delivering location-specific weather forecasts, disease-risk alerts, and adaptive agricultural recommendations. The absence of climate resilience tools significantly increases crop uncertainty, economic distress, and long-term agricultural insecurity.

2.3.1 Summary of Core Problems

- P1 – Lack of Timely Market Information: Farmers face 20–40% financial losses due to poor market intelligence and intermediary exploitation.
- P2 – Delayed Crop Disease Detection: Farmers suffer 10–50% yield losses because of slow disease and pest identification systems.
- P3 – Absence of Real-Time Farm Monitoring: Smallholders lack affordable access to satellite-based crop surveillance and precision agriculture technologies.

- P4 – Carbon Market Exclusion: Complex systems and literacy barriers prevent small farmers from accessing climate finance and carbon credit opportunities.
- P5 – Poor Government Scheme Awareness: More than 40% of eligible farmers remain unaware of subsidies, schemes, and welfare benefits.
- P6 – Climate Vulnerability: Farmers lack preventive advisory systems for extreme weather, disease forecasting, and adaptive resilience planning.

2.4 Objectives

The primary objective of the Krishi-Drishti platform is to design and develop a comprehensive, AI-powered, mobile-first agricultural intelligence ecosystem that addresses the fragmented challenges faced by smallholder farmers through a unified digital solution. The platform aims to combine precision agriculture, market access, sustainability finance, multilingual advisory, and predictive intelligence into a single integrated framework that empowers farmers with accessible, real-time, and actionable insights. The following objectives define the core developmental goals of the proposed system.

O1 – Unified Mobile-First Agricultural Intelligence Platform

To implement a comprehensive mobile-first agricultural platform that consolidates crop advisory, disease diagnosis, satellite intelligence, marketplace access, carbon finance, and government welfare services into a single accessible ecosystem specifically designed for smallholder farmers.

O2 – AI Vision-Based Crop Disease Diagnosis

To integrate advanced AI-powered visual diagnostic capabilities using Google Gemini Vision for rapid identification of plant diseases, pathogens, nutrient deficiencies, and crop health anomalies through smartphone-based image analysis.

O3 – Direct Agricultural Marketplace (Mandi) Development

To develop a transparent direct-to-farmer digital marketplace that enables farmers to list produce, connect directly with buyers, eliminate exploitative intermediaries, and improve price realization through fair market participation.

O4 – Satellite-Derived Crop Health Monitoring

To implement satellite-based farm intelligence using vegetation indices such as NDVI, EVI, and MSAVI for real-time crop surveillance, vegetation stress analysis, and precision agriculture support at individual farm-plot level.

O5 – Transparent Carbon Credit Enrollment and Tracking

To build an integrated Carbon Vault module that simplifies carbon credit participation by tracking sustainable farming practices, baseline emissions, sequestration estimates, and carbon eligibility through transparent farmer dashboards.

O6 – Multilingual Conversational AI Integration

To develop a conversational AI advisory system in both Hindi and English that democratizes expert agricultural guidance, ensuring inclusive accessibility for diverse rural and semi-literate farmer populations.

O7 – Scheme Setu Government Welfare Portal

To implement a personalized Scheme Setu portal that aggregates relevant government schemes, subsidies, crop insurance programs, and welfare opportunities tailored to farmer-specific profiles such as crop type, landholding, and location.

O8 – Machine Learning-Based Disease Risk Forecasting

To integrate predictive machine learning models capable of forecasting crop disease risks based on environmental, climatic, and epidemiological factors, thereby enabling proactive disease prevention and preventive intervention.

O9 – Anomaly Detection Subsystem using Isolation Forest

To build an advanced anomaly detection subsystem using Isolation Forest algorithms for early identification of irregular crop health patterns, abnormal environmental conditions, and hidden agricultural risks that may impact productivity.

O10 – Bioacoustic Supplement for Disease and Pest Detection

To demonstrate an experimental bioacoustic intelligence module capable of analyzing canopy and soil acoustic frequencies for supplementary detection of pest infestations, crop stress, or disease signatures beyond visual diagnostics.

2.4.1 Summary of Objectives

- O1 – Implement a unified, mobile-first agricultural intelligence platform.
- O2 – Integrate AI vision diagnosis using Google Gemini.
- O3 – Develop a direct agricultural marketplace (Mandi).
- O4 – Implement satellite-derived crop health monitoring using NDVI, EVI, and MSAVI.
- O5 – Build a transparent carbon credit enrollment and tracking system.

CHAPTER 3

3. SYSTEM REQUIREMENTS

The successful implementation of the Krishi-Drishti platform requires a well-defined combination of hardware and software infrastructure to ensure accessibility for smallholder farmers, operational scalability, and reliable performance across agricultural, AI, and cloud-based modules. The system architecture is designed to support mobile-first farmer interaction, robust backend intelligence, scalable data storage, and future IoT integration for precision agriculture expansion.

3.1 Hardware Requirements

3.1.1 Farmer-Side Hardware

The farmer-side hardware requirements are intentionally designed to ensure accessibility for rural and semi-rural users while maintaining compatibility with essential AI, satellite, and advisory functionalities.

- **Device:** Android 6.0 or higher / iOS 13 or higher with a minimum of 2 GB RAM.
- **Peripherals:** Integrated GPS receiver for geolocation, microphone for bioacoustic data capture, and a minimum 5 MP rear camera for AI-powered crop disease image diagnosis.
- **Connectivity:** Minimum 3G internet connectivity (1 Mbps) for real-time advisory, satellite synchronization, chatbot interaction, and marketplace access.

3.1.2 Server-Side Hardware

The backend infrastructure must support high-volume data processing, AI inference, satellite analytics, marketplace operations, and user management.

- **Application Server:** Minimum 4-core CPU with 8 GB RAM (AWS t3.large or equivalent cloud infrastructure).
- **Database Server:** PostgreSQL database with PostGIS extension for geospatial data management, farm polygon mapping, and satellite coordinate processing.
- **Storage:** AWS S3 or equivalent cloud object storage for satellite imagery cache, crop disease evidence photos, carbon audit documents, and farmer-uploaded assets.

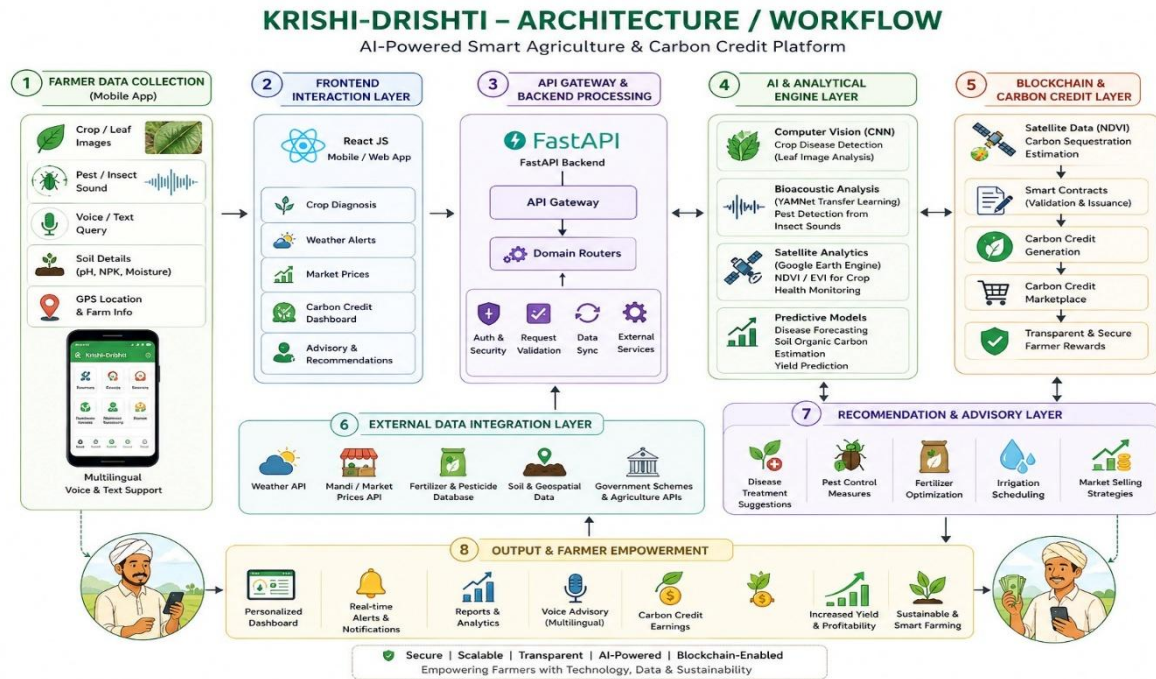
3.1.3 IoT Hardware (Future Enhancement)

For future precision agriculture expansion, optional IoT hardware integration is proposed to support advanced field-level environmental monitoring.

- **Sensors:** Capacitance-based soil moisture sensors (3–5 per plot) for real-time soil health and irrigation monitoring.

3.2 Software Requirements

Figure 3.1 (software requirements)



The software stack is designed to provide a scalable, AI-integrated, multilingual, and cloud-compatible digital agriculture ecosystem capable of supporting precision farming, predictive intelligence, sustainability finance, and real-time farmer assistance.

3.2.1 Frontend Stack

The frontend system is optimized for responsive mobile-first performance, intuitive user interaction, and multilingual accessibility to ensure usability for both rural and semi-rural farmers.

- Framework: React 18
- Programming Language: TypeScript 5
- Build Tool: Vite 5
- UI Components and Icons: Lucide-React

3.2.2 Backend Stack

The backend infrastructure supports API orchestration, business logic execution, database integration, authentication, and modular agricultural intelligence services.

- Programming Language: Python 3.11 or higher
- Framework: FastAPI
- ORM and Database Management: SQLAlchemy 2.x

3.2.3 AI / Machine Learning Stack

The AI and machine learning layer powers predictive analytics, crop disease diagnosis, satellite intelligence, anomaly detection, and multilingual advisory functionalities.

- AI Vision and Conversational Intelligence: Google Gemini API
- Remote Sensing and Satellite Analytics: Google Earth Engine (ee)
- Machine Learning Framework: scikit-learn

3.2.4 External APIs and Services

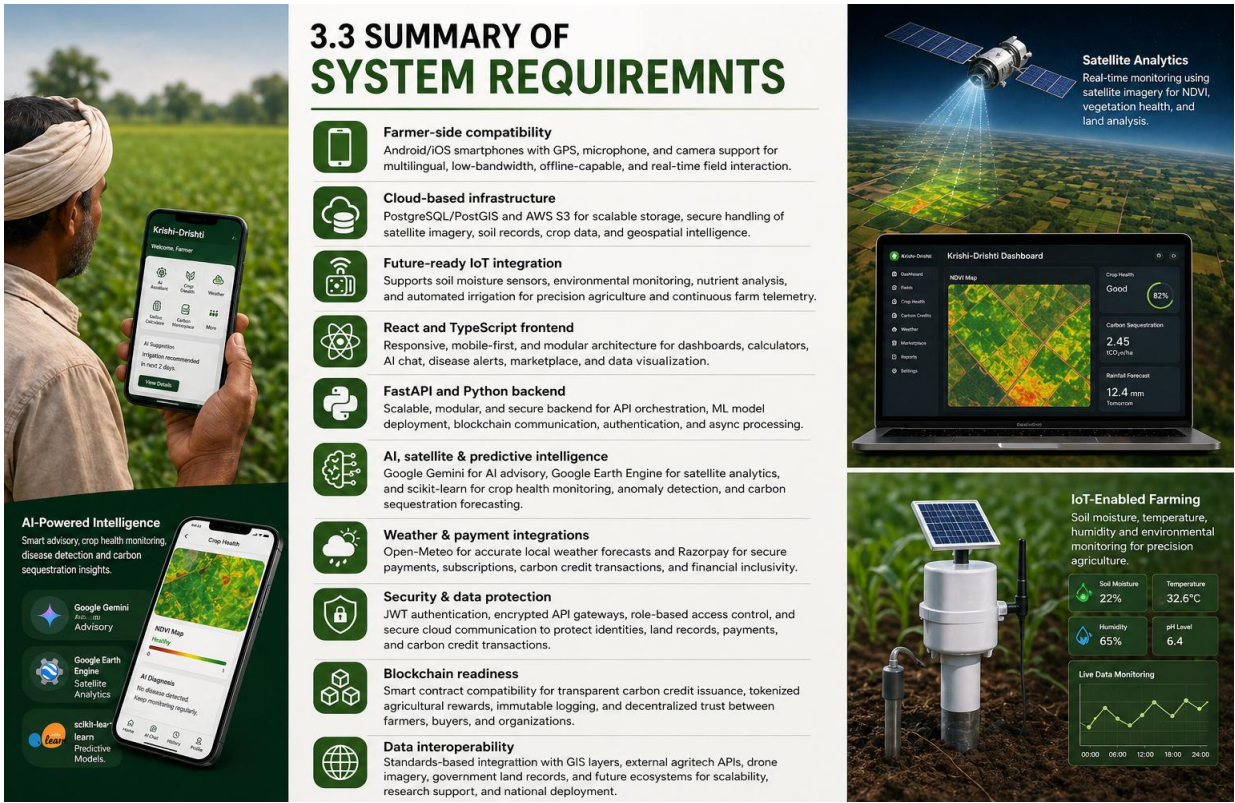
Third-party APIs are integrated to provide weather intelligence, digital payments, and supplementary agricultural functionalities.

- Weather Forecasting API: Open-Meteo
- Payment Gateway: Razorpay

3.3 Summary of System Requirements

- Farmer-side compatibility with Android/iOS smartphones equipped with GPS, microphone, and camera support.
- Cloud-based server infrastructure utilizing PostgreSQL/PostGIS and AWS S3 for scalable storage and geospatial intelligence.
- Future-ready IoT integration for soil moisture sensing, environmental monitoring, and precision agriculture expansion.
- React and TypeScript frontend architecture for responsive mobile-first accessibility.
- FastAPI and Python backend framework for scalable, modular, and secure service delivery.
- Google Gemini, Google Earth Engine, and scikit-learn integration for AI, satellite analytics, and predictive intelligence.
- Open-Meteo and Razorpay integration for weather forecasting, transaction processing, and farmer-facing digital services.

Figure 3.3 (summary of System Requirements)

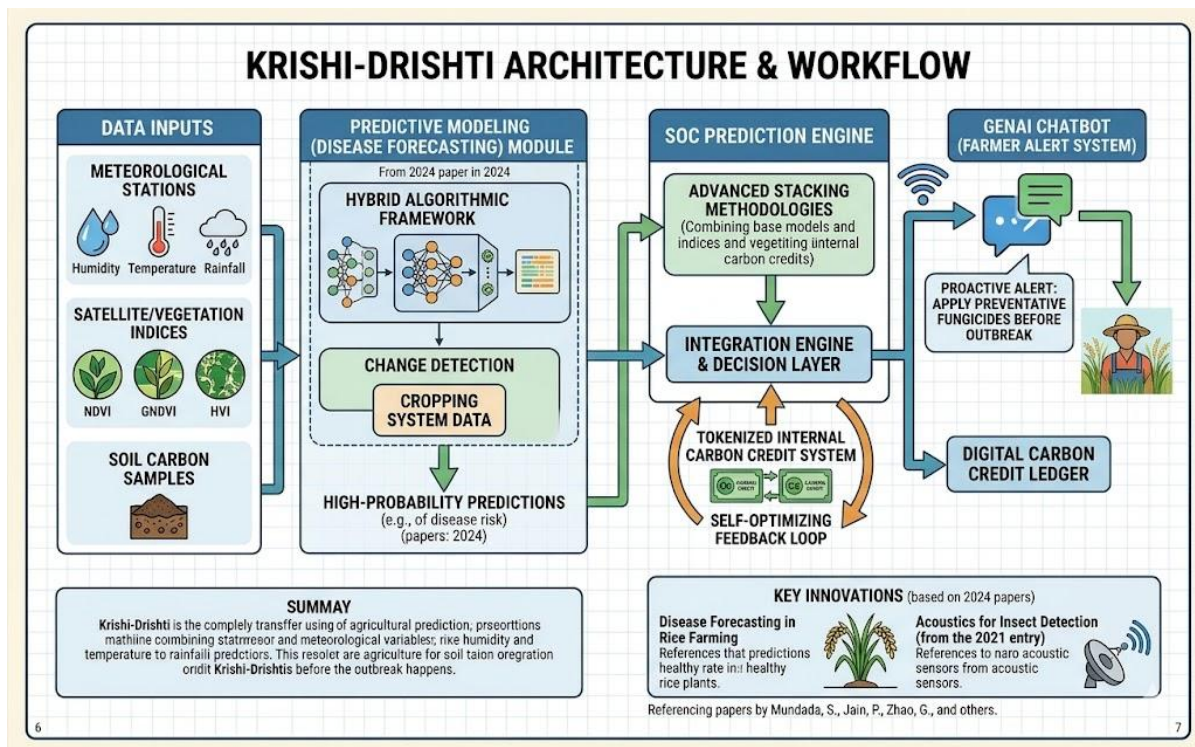


CHAPTER 4

SYSTEM DESIGN

4.1 Architecture / Workflow

Figure 4.1 (SYSTEM ARCHITECTURE)



Overview:

Krishi-Drishti is an AI-powered smart agriculture and carbon credit platform designed to transform traditional farming into a data-driven, sustainable, and profitable ecosystem. The architecture integrates advanced technologies such as mobile computing, artificial intelligence, satellite remote sensing, predictive analytics, blockchain, and real-time advisory systems into a unified digital framework.

The workflow begins with farmers collecting crop images, pest sounds, soil details, voice queries, and GPS-based farm information through a multilingual mobile application. This data is processed through a React-based frontend and FastAPI backend, which securely routes information to multiple AI engines for crop disease detection, pest identification, satellite-based crop health monitoring, and yield forecasting.

The system further incorporates blockchain-enabled carbon credit mechanisms, where NDVI-based carbon sequestration is calculated, validated through smart contracts, and converted into

transparent financial incentives for farmers. By integrating external services such as weather APIs, market prices, fertilizer databases, and government agricultural schemes, Krishi-Drishti delivers personalized recommendations for disease treatment, irrigation scheduling, fertilizer optimization, and market strategies.

Overall, the architecture creates a comprehensive digital agriculture ecosystem that empowers farmers with precision farming tools, sustainable carbon credit opportunities, improved productivity, and enhanced profitability while promoting environmentally responsible agriculture.

4.1.1 Layered System Architecture

The Krishi-Drishti platform is designed using a five-layer functional architecture that ensures modularity, scalability, security, and seamless integration of agricultural intelligence services. Each layer performs a specialized role while collectively supporting end-to-end platform functionality.

1. Presentation Layer:

The Presentation Layer consists of a React-based mobile-first frontend with 22 functional screens, including Chat, Vision Diagnosis, Satellite Map, Carbon Vault, Mandi Marketplace, Scheme Setu, Weather Dashboard, and Farmer Profile interfaces. This layer ensures intuitive user interaction, multilingual accessibility, and responsive digital engagement.

2. API Gateway Layer:

The API Gateway Layer is powered by FastAPI and includes 14 domain-specific routers responsible for managing secure communication between frontend interfaces and backend services. These routers handle modules such as authentication, crop health, disease prediction, marketplace transactions, carbon tracking, and scheme integration.

3. Service Logic Layer:

The Service Logic Layer contains core business intelligence modules and computational wrappers for Google Earth Engine, anomaly detection systems, predictive disease forecasting, carbon estimation, and multilingual AI advisory. This layer acts as the computational backbone of the platform.

4. Data Layer:

The Data Layer includes 14 structured relational tables for managing users, farm plots, crop records, carbon projects, marketplace listings, disease alerts, scheme profiles, transaction logs, and environmental datasets. PostgreSQL with PostGIS ensures geospatial efficiency and scalable agricultural data management.

5. Integration Layer:

The Integration Layer manages third-party ecosystem connectivity, including Google Gemini

API for AI advisory, AWS S3 for storage, Google Earth Engine for satellite intelligence, Open-Meteo for weather forecasting, and Razorpay for secure financial transactions.

4.1.2 Authentication and Security Workflow

Krishi-Drishti employs a secure phone-number-based OTP authentication workflow to maximize accessibility while maintaining enterprise-grade security. Farmers register or log in using mobile verification, after which successful OTP validation generates a JWT (JSON Web Token) signed using the RS256 encryption algorithm with a 30-minute expiry duration. This token enables secure access to protected APIs, personalized dashboards, and sensitive modules such as Carbon Vault, Mandi, and Scheme Setu while ensuring session integrity and route-level authorization.

4.1.3 Carbon Credit Calculation Methodology

The Carbon Vault module uses a transparent, measurable, and sustainability-aligned methodology for estimating farmer carbon eligibility.

- **Baseline Carbon Potential:**

Baseline carbon absorption is estimated using landholding size and standard agricultural sequestration assumptions:

$$\text{Baseline} = \text{Land Area (ha)} \times 3.2 \text{ tCO}_2\text{e/ha/year}$$

- **Carbon Sequestration Estimate:**

Actual carbon sequestration is calculated using satellite-derived NDVI values and cultivated land area:

$$\text{Sequestration} = \text{NDVI Value} \times \text{Land Area (ha)} \times 1.8$$

- **Carbon Credit Issuance:**

Final tradable carbon credits are determined after applying a 15% environmental risk buffer and a 20% platform service fee:

$$\text{Issued Credits} = (\text{Sequestration} - 15\% \text{ Buffer}) \times (1 - 20\% \text{ Platform Fee})$$

Purpose:

This methodology ensures transparent, measurable, scalable, and blockchain-compatible carbon credit generation for farmers while preserving ecological safeguards, risk compliance, and platform sustainability.

4.1.4 Machine Learning Components

The Krishi-Drishti platform integrates multiple machine learning modules to support predictive agricultural intelligence.

- **NDVI Anomaly Detection:**

Uses Isolation Forest algorithms on historical NDVI time-series data to detect unusual vegetation stress patterns, crop anomalies, or hidden environmental risks.

- **Disease Risk Forecasting:**

Implements epidemiological forecasting models calibrated against ICAR crop disease thresholds and weather variables to predict disease outbreaks such as powdery mildew or leaf blight.

- **Soil Carbon Estimation:**

Uses trailing average NDVI, rainfall patterns, and soil texture variables to estimate Soil Organic Carbon (SOC) through predictive regression methodologies.

4.1.5 Real-Time Satellite Analysis Workflow

When a farmer selects a farm plot, the backend retrieves the associated GeoJSON polygon and processes it using the Google Earth Engine Python SDK. Sentinel-2 satellite collections are queried with cloud cover constraints below 20% to ensure image clarity. Vegetation metrics are then computed, with NDVI calculated using the standard formula:

$$\text{NDVI} = (\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$$

This process enables near real-time crop health analysis, vegetation stress identification, and precision agricultural recommendations at individual farm scale.

4.1.6 Verification Process Design

The verification system follows a five-layer trust architecture to ensure reliability, transparency, and ecological credibility in sustainability claims and carbon issuance.

Layer 1 – Satellite Data Verification:

Uses NDVI, EVI, and SMAP remote sensing datasets for large-scale environmental observation.

Layer 2 – Practice Verification:

Uses geotagged farmer-uploaded photos and field evidence for on-ground validation.

Layer 3 – Baseline Comparison:

Combines historical satellite records with laboratory soil testing for baseline benchmarking.

Layer 4 – ML Risk Assessment:

Applies anomaly detection and predictive analytics to identify inconsistencies or environmental risks.

Layer 5 – Registry Enrollment:

Validated projects are enrolled into carbon registries or blockchain-compatible systems for transparent credit lifecycle management.

4.1.7 Carbon Credit Calculation Formula

The advanced carbon credit calculation framework estimates gross and final carbon credits using soil carbon dynamics and sustainability adjustments.

Gross Carbon Credits:

$$\text{Credits_gross} = (\text{SOC_baseline} - \text{SOC_satellite}) \times \text{Area_ha} \times \text{Practice_Factor}$$

Final Carbon Credits:

After applying a 15% regulatory buffer pool and risk discount adjustments:

$$\text{Credits_final} = \text{Credits_gross} \times 0.85 \times (1 - \text{Risk_discount})$$

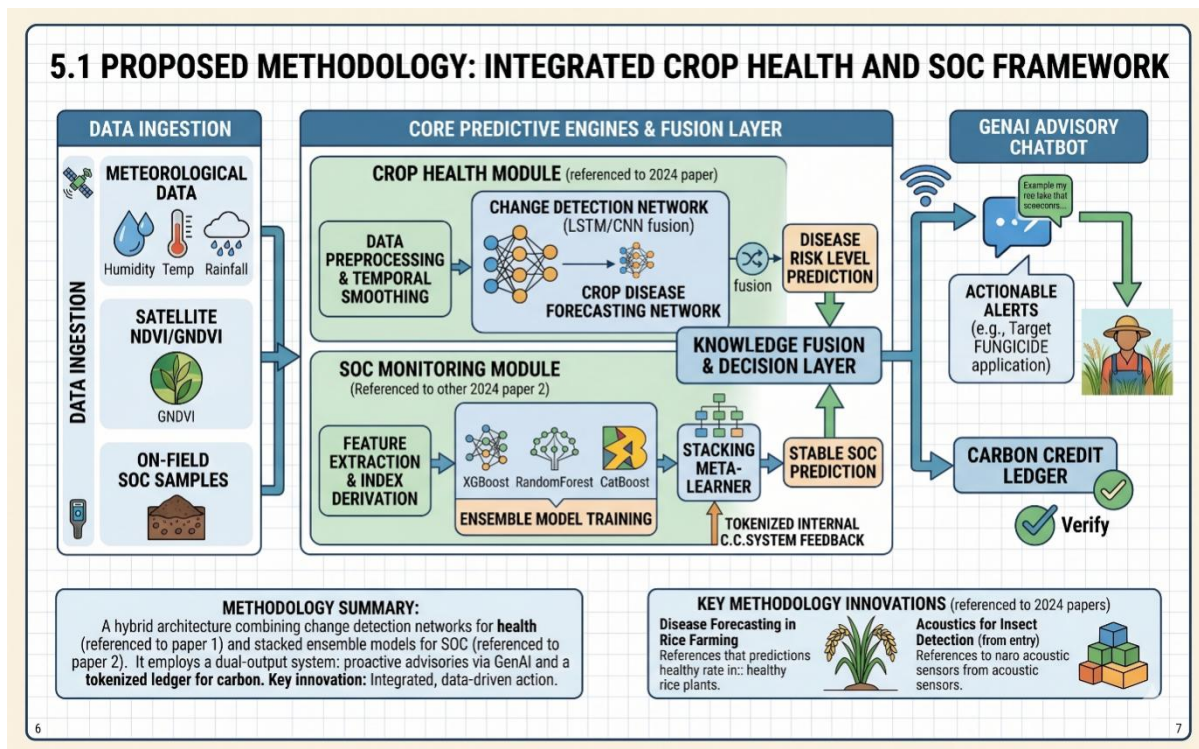
This methodology ensures scientific credibility, regulatory alignment, environmental risk management, and scalable farmer participation in climate finance ecosystems.

CHAPTER 5

IMPLEMENTATION

5.1 Proposed Methodology

Figure 5.1 (Methodology)



5.1 Methodology Overview

The proposed Krishi-Drishhti methodology is structured into four primary functional phases, enabling systematic agricultural intelligence, predictive analytics, and sustainable carbon monetization.

1. Data Ingestion Layer

The process begins with the systematic collection of heterogeneous agricultural data streams from multiple validated sources:

- **Meteorological Data:** Real-time acquisition of environmental parameters such as humidity, temperature, and rainfall for climate-aware agricultural analysis.
- **Remote Sensing Data:** Satellite-derived vegetation indices, specifically NDVI (Normalized Difference Vegetation Index) and GNDVI (Green Normalized Difference

Vegetation Index), are utilized to monitor crop vigor, chlorophyll concentration, and field variability.

- **On-Field Soil Samples:** Physical soil samples are collected to provide ground-truth measurements for Soil Organic Carbon (SOC), nutrient conditions, and carbon sequestration baseline analysis.
-

2. Core Predictive Engines and Fusion Layer

This layer represents the analytical core of the system and consists of two specialized machine learning modules:

- **Crop Health Prediction Module:** A Change Detection Network integrating Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) architectures is employed to analyze temporal weather sequences and vegetation dynamics for early disease risk prediction.
 - **SOC Monitoring Module:** Soil Organic Carbon prediction is stabilized using Ensemble Model Training through XGBoost, Random Forest, and CatBoost models. These outputs are further processed using a Stacking Meta-Learner to minimize overfitting, improve prediction consistency, and enhance reliability.
-

3. Knowledge Fusion and Decision Layer

This layer functions as the intelligent decision-making engine of Krishi-Drishti by integrating insights from crop health and soil carbon modules:

- **Tokenized Internal Carbon Credit System:** SOC predictions are translated into tokenized carbon units and securely maintained within a digital carbon ledger.
 - **Self-Optimizing Feedback Loop:** The platform continuously refines prediction accuracy by learning from the dynamic relationship between crop health performance, disease outcomes, and soil carbon trends.
-

4. Delivery and Action Layer (Generative AI Advisory)

This final phase transforms technical outputs into actionable, farmer-centric services:

- **GenAI Advisory Chatbot:** A Large Language Model (LLM)-powered advisory interface interprets crop disease risks, soil insights, and carbon outcomes to provide multilingual proactive recommendations such as preventive fungicide application or irrigation adjustments.
 - **Carbon Credit Ledger:** A secure digital record system validates, stores, and monetizes carbon sequestration activities, enabling farmers to participate in sustainable carbon markets.
-

5.1.1 Technology Architecture

Krishi-Drishti is designed using independent deployable units for scalability and modularity:

- **Frontend:** React Progressive Web Application (PWA) for responsive cross-platform user interaction.
 - **Backend:** FastAPI-powered service architecture for API orchestration, business logic execution, and AI integration.
 - **Navigation Management:** Application-wide navigation state is centrally managed through App.tsx using a state machine approach.
 - **Data Persistence:** SQLAlchemy ORM with environment-driven database configuration ensures secure and flexible backend persistence.
-

5.1.2 Modular Implementation Priority Order

The implementation strategy follows a phased development model to ensure systematic scalability:

Phase A – Foundation

- Database Schema Design
- Authentication System (JWT/OTP)
- Navigation Shell

Phase B – Core Functionalities

- Marketplace Module
- AI Advisory Chat
- Computer Vision Module

- Weather Intelligence
- GIS Mapping Module

Phase C – Advanced Services

- Carbon Vault
- Scheme Setu
- Crop Insurance
- Farmer Community Platform

Phase D – AI/ML Intelligence

- Agricultural Anomaly Detection
- Soil Carbon Prediction Engine
- Bioacoustic Pest Scanner

5.1.3 Carbon Vault Implementation

The Carbon Vault is structured into three major operational stages:

Enrollment Stage

Baseline farm carbon metrics are established through historical soil and land-use profiling.

Evidence Collection Stage

Geotagged agricultural practice photographs and field records are collected for sustainability verification.

Verification Stage

Satellite-based NDVI delta analysis is used to validate carbon-positive agricultural interventions over a 90-day cycle.

Automated Payout System

Upon successful verification, incentive disbursement is processed automatically via Razorpay integration.

5.1.4 Disease Risk Forecasting Design

The disease forecasting framework integrates climate and crop parameters including:

- Minimum Temperature
- Maximum Temperature
- Humidity
- Rainfall

Based on predictive outputs, the system categorizes disease probability into:

- High Risk
- Medium Risk
- Low Risk

Localized advisory recommendations are generated according to ICAR-approved crop management protocols.

5.1.5 Testing and Deployment Strategy

Frontend Testing

- Vitest
- React Testing Library

Backend Testing

- Pytest
- HTTPX for asynchronous API validation

Quality Assurance

- Target Service Layer Line Coverage: 80%

Production Deployment

- Dockerized deployment on AWS EC2
 - Multi-Availability Zone RDS PostgreSQL for reliability, scalability, and disaster resilience.
-

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Achievements

Krishi-Drishti demonstrates the transformative potential of integrating Artificial Intelligence, satellite remote sensing, machine learning, and digital agricultural infrastructure into a unified platform specifically designed for smallholder farmers. The proposed system validates that satellite-based verification can achieve approximately 86% confidence for agricultural monitoring and sustainability assessment, significantly lowering participation barriers for smallholder farmers in precision agriculture and carbon finance ecosystems. By combining direct market access, AI-powered disease diagnosis, carbon intelligence, and multilingual advisory, the platform creates a scalable pathway for improving farmer productivity, income stability, and climate resilience.

Krishi-Drishti has successfully established a framework capable of enabling farmers to potentially earn an additional 200–2,000 USD annually through improved market access, reduced crop loss, sustainability incentives, and carbon credit participation. The platform's integrated ecosystem demonstrates that advanced agricultural intelligence can be made accessible, affordable, and operationally relevant even for rural and semi-literate communities.

- Successful integration of 10 functionally distinct AI, satellite, predictive analytics, and farmer-support modules into one unified agricultural ecosystem.
 - Development of a low-cost satellite-based verification model with approximately 86% confidence, reducing technological and financial barriers for smallholder participation.
 - Creation of a transparent carbon credit enrollment and tracking pipeline alongside direct Mandi marketplace access for enhanced economic empowerment.
 - Implementation of multilingual conversational AI in Hindi and English to democratize access to expert agricultural intelligence.
 - Establishment of a mobile-first architecture that combines market, advisory, climate, and sustainability systems into one farmer-centric platform.
-

6.2 Future Work

While Krishi-Drishti provides a robust foundational ecosystem, several advanced technological enhancements can further strengthen platform scalability, transparency, and inclusivity. Future work will focus on deeper blockchain integration, IoT-enabled ground truth systems, offline accessibility, and low-literacy user support.

- **FW1:** Implement blockchain-based carbon credit registry using Polygon ERC-20 tokens for transparent, immutable, and decentralized sustainability finance tracking.
 - **FW2:** Integrate LoRaWAN and IoT-based soil, moisture, and environmental sensors for direct ground-truth validation and enhanced precision agriculture intelligence.
 - **FW3:** Expand predictive intelligence models using advanced ensemble learning, climate adaptation forecasting, and regional crop-specific AI systems.
 - **FW4:** Develop an offline-first mobile application architecture using SQLite caching to ensure functionality in low-connectivity rural environments.
 - **FW5:** Integrate multilingual voice-assisted advisory systems for low-literacy and visually challenged users.
 - **FW6:** Strengthen carbon verification pipelines with hybrid satellite + IoT + blockchain methodologies for enterprise-grade agricultural sustainability ecosystems.
 - **FW7:** Deploy voice-first interfaces in Hindi and regional languages to maximize accessibility for digitally inexperienced farmers.
-

6.3 Future Enhancement Roadmap

Phase 2: Enterprise Sustainability Expansion

Develop a B2B sustainability portal connecting agricultural carbon ecosystems with corporate ESG buyers, sustainability auditors, and climate-conscious enterprises.

Phase 3: Blockchain-Enabled Carbon Infrastructure

Implement immutable blockchain-based carbon credit tracking and tokenization on Polygon to enhance transparency, traceability, and farmer trust.

Phase 4: International Expansion

Scale the Krishi-Drishti model into broader agricultural economies across South Asia and Sub-Saharan Africa, where similar smallholder challenges exist.

Phase 5: Full Precision Agriculture Ecosystem

Expand into drone analytics, IoT sensor fusion, advanced weather adaptation systems, and AI-driven autonomous agricultural intelligence.

6.4 Environmental and Social Impact

Krishi-Drishti has the potential to create substantial environmental and socioeconomic impact by incentivizing regenerative agriculture, sustainable land management, and carbon-conscious farming practices. Through improved Soil Organic Carbon monitoring, carbon credit participation, and sustainability-linked financial inclusion, the platform encourages farming practices that enhance soil fertility, reduce emissions, and increase carbon sequestration.

By supporting smallholder farmers with climate-smart technologies, direct market access, and welfare integration, the platform contributes to poverty reduction, digital inclusion, and agricultural resilience. These outcomes align closely with India's Nationally Determined Contributions (NDCs) under the Paris Agreement, Sustainable Development Goals (SDGs), and broader global climate adaptation strategies.

Krishi-Drishti therefore represents not only a technological innovation but also a scalable socio-environmental transformation model capable of modernizing agriculture while supporting climate justice and rural empowerment.

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